

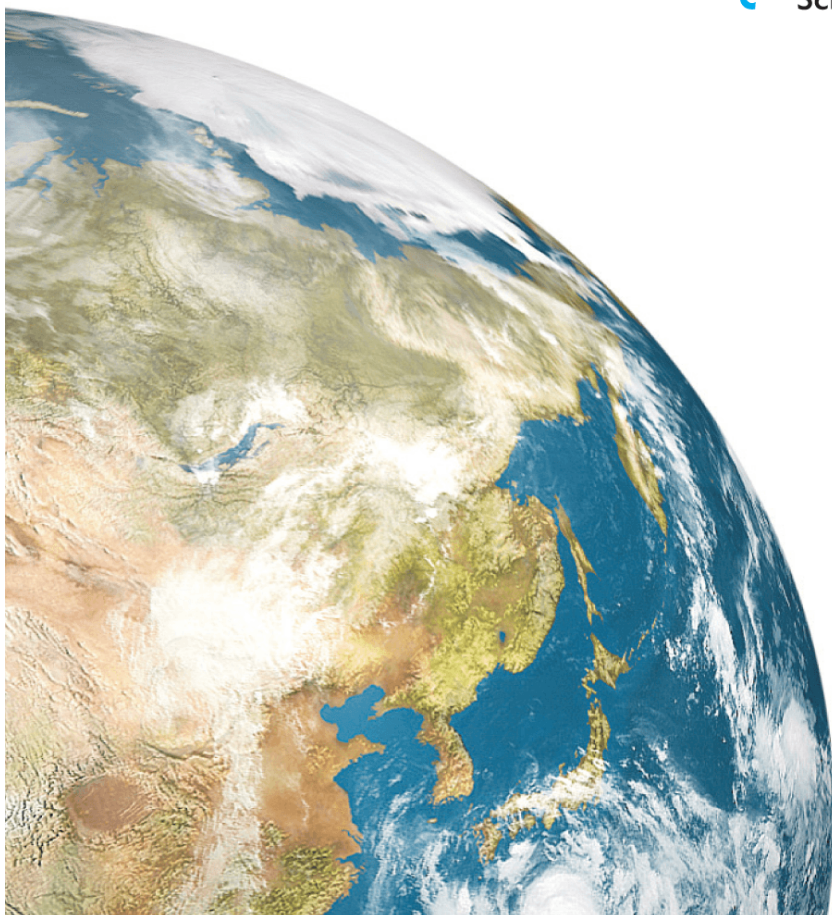
Chapter

1

Indices

Contents:

- A** Index laws
- B** Rational indices
- C** Scientific notation (standard form)



Example 1
Self Tutor

Express in simplest form with a prime number base:

a 9^4

b 4×2^p

c $\frac{3^x}{9y}$

d 25^{x-1}

$$\begin{aligned} \mathbf{a} \quad 9^4 &= (3^2)^4 \\ &= 3^{2 \times 4} \\ &= 3^8 \end{aligned}$$

$$\begin{aligned} \mathbf{b} \quad 4 \times 2^p &= 2^2 \times 2^p \\ &= 2^{2+p} \end{aligned}$$

$$\begin{aligned} \mathbf{c} \quad \frac{3^x}{9y} &= \frac{3^x}{(3^2)^y} \\ &= \frac{3^x}{3^{2y}} \\ &= 3^{x-2y} \end{aligned}$$

$$\begin{aligned} \mathbf{d} \quad 25^{x-1} &= (5^2)^{x-1} \\ &= 5^{2(x-1)} \\ &= 5^{2x-2} \end{aligned}$$

EXERCISE 1A
1 Simplify using the index laws:

a $3^2 \times 3^5$

b $x^6 \times x^3$

c $x^5 \times x^n$

d $t^3 \times t^4 \times t^5$

e $\frac{7^9}{7^5}$

f $\frac{x^7}{x^3}$

g $\frac{t^6}{t^x}$

h $t^{3m} \div t$

i $(5^3)^2$

j $(t^4)^3$

k $(y^3)^m$

l $(a^{3m})^4$

2 Express in simplest form with a prime number base:

a 121

b 32

c 81

d 4^2

e 25^2

f $7^t \times 49$

g $3^a \div 9$

h $8^p \div 4$

i $\frac{7^n}{7^{n-2}}$

j $\frac{9}{3^x}$

k $(25^t)^2$

l $16^{k-3} \times 2^{-k}$

m $\frac{4^a}{2^b}$

n $\frac{8^x}{16^y}$

o $\frac{125^{x+1}}{5^{x-1}}$

p $\frac{27^{a+2}}{3^a \times 9^a}$

Example 2
Self Tutor

Remove the brackets of:

a $(2x)^3$

b $\left(\frac{3c}{b}\right)^4$

$$\begin{aligned} \mathbf{a} \quad (2x)^3 &= 2^3 \times x^3 \\ &= 8x^3 \end{aligned}$$

$$\begin{aligned} \mathbf{b} \quad \left(\frac{3c}{b}\right)^4 &= \frac{3^4 \times c^4}{b^4} \\ &= \frac{81c^4}{b^4} \end{aligned}$$

Each factor within the brackets is raised to the power outside them.


3 Remove the brackets of:

a $(xy)^2$

b $(ab)^3$

c $(xyz)^2$

d $(3b)^3$

e $(5a)^4$

f $(10xy)^5$

g $\left(\frac{p}{q}\right)^2$

h $\left(\frac{m}{n}\right)^3$

OPENING PROBLEM

Approximately 58 400 000 vehicles cross the Sydney Harbour Bridge each year.

Things to think about:

- a Can you write this number in the form $a \times 10^k$ where $1 \leq a < 10$ and k is an integer?
- b Can you use the number in this form to estimate how many vehicles cross the bridge:
 - i each day
 - ii over 10 years?



INDEX NOTATION

We often deal with numbers that are repeatedly multiplied together, such as $5 \times 5 \times 5$. We can use **indices** or **exponents** to conveniently represent such expressions.

Using **index notation**, we represent $5 \times 5 \times 5$ as 5^3 , which reads “5 to the power 3”. We say that 5 is the **base**, and 3 is the **index** or **power** or **exponent**.

If n is a positive integer, then a^n is the product of n factors of a .

$$a^n = \underbrace{a \times a \times a \times a \times \dots \times a}_{n \text{ factors}}$$

A

INDEX LAWS

In previous years we have seen the following **index laws**:

If the bases a and b are both positive, and the indices m and n are integers, then:

$$a^m \times a^n = a^{m+n}$$

To **multiply** numbers with the **same base**, keep the base and **add** the indices.

$$\frac{a^m}{a^n} = a^{m-n}$$

To **divide** numbers with the same base, keep the base and **subtract** the indices.

$$(a^m)^n = a^{mn}$$

When **raising** a **power** to a **power**, keep the base and **multiply** the indices.

$$(ab)^n = a^n b^n$$

The power of a product is the product of the powers.

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

The power of a quotient is the quotient of the powers.

$$a^0 = 1, \quad a \neq 0$$

Any non-zero number raised to the power of zero is 1.

$$a^{-n} = \frac{1}{a^n}$$

and in particular, $a^{-1} = \frac{1}{a}$.

i $\left(\frac{x}{3}\right)^4$

j $\left(\frac{5}{z}\right)^3$

k $\left(\frac{2a}{b}\right)^4$

l $\left(\frac{3x}{4y}\right)^3$

4 Simplify the following expressions using one or more of the index laws:

a $4b^2 \times 2b^3$

b $\frac{a^6b^3}{a^4b}$

c $3ab^2 \times 2a^3$

d $\frac{5x^3y^2}{15xy}$

e $\left(\frac{a^2}{5b}\right)^3$

f $\frac{24t^6r^4}{15t^6r^2}$

g $\frac{(4c^3d^2)^2}{c^2d}$

h $\frac{10k^7}{(2k)^5}$

Example 3**Self Tutor**

Simplify, giving your answers in simplest rational form:

a 7^0

b 3^{-2}

c $3^0 - 3^{-1}$

d $\left(\frac{5}{3}\right)^{-2}$

a $7^0 = 1$

b $3^{-2} = \frac{1}{3^2} = \frac{1}{9}$

c $3^0 - 3^{-1} = 1 - \frac{1}{3} = \frac{2}{3}$

d $\left(\frac{5}{3}\right)^{-2} = \left(\frac{3}{5}\right)^2 = \frac{9}{25}$

Notice that

$$\left(\frac{a}{b}\right)^{-2} = \left(\frac{b}{a}\right)^2$$



5 Simplify, giving your answers in simplest rational form:

a 3^0

b 6^{-1}

c 4^{-1}

d 5^0

e 4^2

f 4^{-2}

g 5^3

h 5^{-3}

i 7^2

j 7^{-2}

k 10^3

l 10^{-3}

6 Simplify, giving your answers in simplest rational form:

a $\left(\frac{1}{2}\right)^0$

b $\frac{5^4}{5^4}$

c $2t^0$

d $(2t)^0$

e 7^0

f 3×4^0

g $\frac{5^3}{5^5}$

h $\frac{2^6}{2^{10}}$

i $\frac{x^4}{x^9}$

j $\left(\frac{3}{8}\right)^{-1}$

k $\left(\frac{2}{3}\right)^{-1}$

l $\left(\frac{1}{5}\right)^{-1}$

m $2^0 + 2^1$

n $5^0 - 5^{-1}$

o $3^0 + 3^1 - 3^{-1}$

p $\left(\frac{1}{3}\right)^{-2}$

q $\left(\frac{2}{3}\right)^{-3}$

r $\left(1\frac{1}{2}\right)^{-3}$

s $\left(\frac{4}{5}\right)^{-2}$

t $\left(2\frac{1}{2}\right)^{-2}$

7 Write the following without brackets or negative indices:

a $(3b)^{-1}$

b $3b^{-1}$

c $7a^{-1}$

d $(7a)^{-1}$

e $\left(\frac{1}{t}\right)^{-2}$

f $\left(\frac{3x}{y}\right)^{-1}$

g $(5t)^{-2}$

h $(5t^{-2})^{-1}$

i xy^{-1}

j $(xy)^{-1}$

k xy^{-3}

l $(xy)^{-3}$

m $(3pq)^{-1}$

n $3(pq)^{-1}$

o $3pq^{-1}$

p $\frac{(xy)^3}{y^{-2}}$

q $(5x^{-2}y^3)^3$

r $\left(\frac{c}{2d^3}\right)^{-2}$

s $\left(\frac{3r^{-3}}{t}\right)^{-2}$

t $\left(\frac{2p}{5q^{-2}}\right)^{-3}$

8 Use the index laws to show that, for positive a and b , and integer n :

a $\frac{1}{a^{-n}} = a^n$

b $\left(\frac{a}{b}\right)^{-n} = \frac{b^n}{a^n}$

9 The units for speed *kilometres per hour* can be written as km/h or km h^{-1} .

Write these units in index form:

a m/s

b cubic metres/hour

c square centimetres per second

d cubic centimetres per minute

e grams per second

f kilogram metres per second

g metres per second per second.

10 Find the smaller of 2^{125} and 3^{75} without a calculator.

Hint: $2^{125} = (2^5)^{25}$

11 Order the following numbers from smallest to largest: 2^{90} , 3^{60} , 5^{36} , 10^{24} .

B

RATIONAL INDICES

A **rational** number is a number which can be written in the form $\frac{p}{q}$ where p and q are integers. The integers themselves are rational numbers, since for example $2 = \frac{2}{1}$.

The index laws can be applied not just to integer indices, but to rational indices in general. This helps to give meaning to values such as $5^{\frac{1}{2}}$ and $7^{\frac{1}{3}}$.

In previous years we have established that: $a^{\frac{1}{2}} = \sqrt{a}$ and $a^{\frac{1}{3}} = \sqrt[3]{a}$.

In general, $a^{\frac{1}{n}} = \sqrt[n]{a}$ where $\sqrt[n]{a}$ is called the **n th root of a** .

Example 4

Self Tutor

Simplify:

a $49^{\frac{1}{2}}$

b $27^{\frac{1}{3}}$

c $49^{-\frac{1}{2}}$

d $27^{-\frac{1}{3}}$

a $49^{\frac{1}{2}}$
 $= \sqrt{49}$
 $= 7$

b $27^{\frac{1}{3}}$
 $= \sqrt[3]{27}$
 $= 3$

c $49^{-\frac{1}{2}}$
 $= \frac{1}{49^{\frac{1}{2}}}$
 $= \frac{1}{\sqrt{49}}$
 $= \frac{1}{7}$

d $27^{-\frac{1}{3}}$
 $= \frac{1}{27^{\frac{1}{3}}}$
 $= \frac{1}{\sqrt[3]{27}}$
 $= \frac{1}{3}$

INVESTIGATION 1**RATIONAL INDICES**

This Investigation will help you discover the meaning of numbers raised to rational indices of the form $\frac{m}{n}$ where $m, n \in \mathbb{Z}$, $n \neq 0$.

For example, what does $8^{\frac{2}{3}}$ mean?

What to do:

- 1 Use the rule $(a^m)^n = a^{mn}$ to simplify $(8^2)^{\frac{1}{3}}$ and $(8^{\frac{1}{3}})^2$.
- 2 Simplify: **a** $(8^2)^{\frac{1}{3}} = \sqrt[3]{8^2} = \dots\dots$ **b** $(8^{\frac{1}{3}})^2 = (\sqrt[3]{8})^2 = \dots\dots$
- 3 Hence write $a^{\frac{m}{n}}$ in two different forms.

From the **Investigation** you should have discovered that

$$a^{\frac{m}{n}} = (\sqrt[n]{a})^m = \sqrt[n]{a^m}.$$

When dealing with indices of this form, it is often easiest to write the base number as a prime raised to a power. We simplify the result using the index laws.

Example 5**Self Tutor**

Evaluate without using a calculator:

a $8^{\frac{4}{3}}$

b $32^{-\frac{2}{5}}$

$$\begin{aligned} \mathbf{a} \quad 8^{\frac{4}{3}} &= (2^3)^{\frac{4}{3}} \\ &= 2^{3 \times \frac{4}{3}} \\ &= 2^4 \\ &= 16 \end{aligned}$$

$$\begin{aligned} \mathbf{b} \quad 32^{-\frac{2}{5}} &= (2^5)^{-\frac{2}{5}} \\ &= 2^{5 \times -\frac{2}{5}} \\ &= 2^{-2} \\ &= \frac{1}{4} \end{aligned}$$

EXERCISE 1B

- 1 Evaluate without using your calculator:

a $16^{\frac{1}{2}}$

b $16^{-\frac{1}{2}}$

c $25^{\frac{1}{2}}$

d $25^{-\frac{1}{2}}$

e $8^{\frac{1}{3}}$

f $8^{-\frac{1}{3}}$

g $(-8)^{\frac{1}{3}}$

h $(-8)^{-\frac{1}{3}}$

i $81^{\frac{1}{4}}$

j $81^{-\frac{1}{4}}$

k $32^{\frac{1}{5}}$

l $32^{-\frac{1}{5}}$

- 2 Evaluate if possible:

a $(-1)^{\frac{1}{2}}$

b $(-1)^{\frac{1}{3}}$

c $(-27)^{-\frac{1}{3}}$

d $(-64)^{-\frac{1}{2}}$

3 Write the following in index form:

a $\sqrt{10}$

b $\frac{1}{\sqrt{10}}$

c $\sqrt[3]{15}$

d $\frac{1}{\sqrt[3]{15}}$

e $\sqrt[4]{19}$

f $\frac{1}{\sqrt[4]{19}}$

g $\sqrt[5]{13}$

h $\frac{1}{\sqrt[5]{13}}$

4 Evaluate without using a calculator:

a $8^{\frac{2}{3}}$

b $4^{\frac{3}{2}}$

c $4^{\frac{5}{2}}$

d $8^{\frac{5}{3}}$

e $16^{\frac{3}{4}}$

f $9^{\frac{3}{2}}$

g $9^{-\frac{3}{2}}$

h $4^{-\frac{1}{2}}$

i $32^{\frac{1}{5}}$

j $32^{\frac{2}{5}}$

k $32^{\frac{3}{5}}$

l $16^{-\frac{3}{4}}$

m $8^{-\frac{2}{3}}$

n $27^{-\frac{4}{3}}$

o $25^{-\frac{3}{2}}$

5 Write the following as powers of 2:

a $\sqrt{8}$

b $\sqrt[3]{32}$

c $\sqrt[4]{4}$

d $\sqrt[3]{16}$

e $\frac{1}{\sqrt[4]{8}}$

f $\frac{1}{\sqrt[3]{16}}$

g $\frac{1}{\sqrt[7]{8}}$

h $\frac{1}{\sqrt[5]{64}}$

i $8\sqrt{2}$

j $4\sqrt{32}$

k $\frac{2}{\sqrt[3]{4}}$

l $\frac{4\sqrt{32}}{8}$

6 Write the following as powers of 3:

a $\sqrt[3]{9}$

b $\sqrt{27}$

c $\frac{1}{\sqrt[4]{27}}$

d $\frac{1}{\sqrt[5]{81}}$

e $9\sqrt{3}$

f $3\sqrt{27}$

g $\frac{9}{\sqrt[5]{3}}$

h $\frac{\sqrt[3]{81}}{9}$

7 Write with a prime number base:

a $\sqrt[3]{25}$

b $\sqrt[4]{32}$

c $\sqrt[5]{125}$

d $\sqrt[7]{121}$

e $\frac{1}{\sqrt[3]{49}}$

f $\sqrt[5]{64}$

g $\frac{1}{\sqrt[7]{625}}$

h $\frac{1}{\sqrt[6]{243}}$

i $16\sqrt{8}$

j $25\sqrt{125}$

k $\frac{13}{\sqrt[3]{169}}$

l $\frac{81}{\sqrt{27}}$

8 Use your calculator to evaluate, rounded to 3 significant figures where necessary:

a $25^{\frac{3}{2}}$

b $27^{\frac{2}{3}}$

c $8^{\frac{7}{3}}$

d $9^{\frac{2}{5}}$

e $10^{\frac{3}{7}}$

f $15^{\frac{5}{3}}$

g $10^{\frac{2}{7}}$

h $18^{\frac{7}{3}}$

i $16^{\frac{3}{11}}$

j $146^{\frac{4}{9}}$

k $4^{-\frac{5}{2}}$

l $27^{-\frac{5}{3}}$

m $15^{-\frac{2}{5}}$

n $53^{-\frac{3}{7}}$

o $3^{-\frac{7}{5}}$

9 Without using your calculator, evaluate $\frac{\sqrt[3]{9} \times \sqrt[4]{27}}{\sqrt[12]{243}}$.

GAME

This game can be played by 2 players.

$\frac{1}{4}$	5	3	$\frac{1}{3}$	2	4	$\frac{1}{8}$	2	$\frac{4}{5}$
2	9	4	$\frac{2}{5}$	$\frac{2}{3}$	81	3	$-\frac{1}{2}$	81
32	-3	$\frac{1}{27}$	125	$\frac{1}{16}$	64	$\frac{1}{2}$	243	5
4	3	$\frac{1}{8}$	-2	3	32	7	$\frac{3}{4}$	2
5	0	2	$\frac{1}{2}$	$-\frac{2}{3}$	1	6	2	36
49	25	$\frac{1}{2}$	-3	0	-2	64	7	$\frac{1}{5}$
25	$\frac{1}{81}$	-4	27	2	-3	5	4	27
-1	6	$-\frac{1}{3}$	16	-2	$\frac{1}{16}$	3	125	$\frac{1}{2}$
$\frac{1}{25}$	16	343	$\frac{1}{2}$	5	3	1	9	$\frac{1}{5}$

PRINTABLE
BOARD



What to do:

- Taking alternate turns, each player selects 3 squares on the board to create a statement of the form $a^b = c$.
For example, the shaded squares can be used to create the statement $3^4 = 81$. These squares are then crossed out and cannot be used again.
- The last player who is able to make a valid selection is the winner.

Single player variant:

- Try to use all 81 squares in 27 selections.

C

SCIENTIFIC NOTATION (STANDARD FORM)

Observe the pattern:

$$\begin{array}{l}
 \div 10 \quad 10\,000 = 10^4 \quad -1 \\
 \div 10 \quad 1000 = 10^3 \quad -1 \\
 \div 10 \quad 100 = 10^2 \quad -1 \\
 \div 10 \quad 10 = 10^1 \quad -1 \\
 \div 10 \quad 1 = 10^0 \quad -1 \\
 \div 10 \quad \frac{1}{10} = 10^{-1} \quad -1 \\
 \div 10 \quad \frac{1}{100} = 10^{-2} \quad -1 \\
 \div 10 \quad \frac{1}{1000} = 10^{-3} \quad -1
 \end{array}$$

As we divide by 10,
the **exponent** or **power**
of 10 decreases by one.



We can use this pattern to simplify the writing of very large and very small numbers.

For example,

$$\begin{aligned} 5\,000\,000 &= 5 \times 1\,000\,000 \\ &= 5 \times 10^6 \end{aligned} \quad \text{and} \quad \begin{aligned} 0.000\,003 &= \frac{3}{1\,000\,000} \\ &= 3 \times \frac{1}{1\,000\,000} \\ &= 3 \times 10^{-6} \end{aligned}$$

Scientific notation involves writing any given number as *a number between 1 inclusive and 10, multiplied by a power of 10*. The result has the form

$$a \times 10^k \quad \text{where } 1 \leq a < 10 \quad \text{and } k \text{ is an integer.}$$

Example 6



Write in scientific notation:

a 23 600 000 **b** 0.000 023 6

a $\overbrace{23\,600\,000}^{7 \text{ digits}}$
 $= 2.36 \times 10^7$

b $\overbrace{0.000\,023\,6}^{5 \text{ digits after decimal}}$
 $= 2.36 \times 10^{-5}$

Example 7



Write as an ordinary decimal number:

a 2.57×10^4 **b** 7.853×10^{-3}

a 2.57×10^4
 $= \overbrace{2.5700}^{4 \text{ digits after decimal}} \times 10\,000$
 $= 25\,700$

b 7.853×10^{-3}
 $= \overbrace{0007.853}^{3 \text{ digits before decimal}} \div 10^3$
 $= 0.007\,853$

Remember that
 $10^{-3} = \frac{1}{10^3}$.



EXERCISE 1C

1 Write using scientific notation:

a 230 **b** 53 900 **c** 0.0361 **d** 0.006 80
e 3.26 **f** 0.5821 **g** 361 000 000 **h** 0.000 001 674

2 Write as an ordinary decimal number:

a 2.3×10^3 **b** 2.3×10^{-2} **c** 5.64×10^5 **d** 7.931×10^{-4}
e 9.97×10^0 **f** 6.04×10^7 **g** 4.215×10^{-1} **h** 3.621×10^{-8}

3 Express the following quantities using scientific notation:

- a** There are approximately 4 million red blood cells in a drop of blood.
b The thickness of a coin is about 0.0008 m.
c Earth's radius is about 6.38 million metres.
d A Rubik's Cube has approximately 43 252 000 000 000 000 000 possible arrangements.



- 4 Express the following quantities as ordinary decimal numbers:

- a The Amazon River is approximately 6.99×10^6 m long.
- b A piece of paper is about 1.8×10^{-2} cm thick.
- c A test tube holds 3.2×10^7 bacteria.
- d A mushroom weighs 8.2×10^{-6} tonnes.

**Example 8****Self Tutor**

Simplify, writing your answer in scientific notation:

a $(3 \times 10^4) \times (8 \times 10^3)$

b $\frac{2 \times 10^{-3}}{5 \times 10^{-8}}$

a $(3 \times 10^4) \times (8 \times 10^3)$
 $= 24 \times 10^{4+3}$
 $= (2.4 \times 10^1) \times 10^7$
 $= 2.4 \times 10^8$

b $\frac{2 \times 10^{-3}}{5 \times 10^{-8}}$
 $= \frac{2}{5} \times 10^{-3-(-8)}$
 $= 0.4 \times 10^5$
 $= (4 \times 10^{-1}) \times 10^5$
 $= 4 \times 10^4$

- 5 Simplify the following, writing your answers in scientific notation:

- a $(3 \times 10^3) \times (2 \times 10^7)$
- b $(4 \times 10^3) \times (7 \times 10^5)$
- c $(8 \times 10^{-4}) \times (7 \times 10^{-5})$
- d $(9 \times 10^{-5}) \times (6 \times 10^{-2})$
- e $(3 \times 10^5)^2$
- f $(4 \times 10^7)^2$
- g $(2 \times 10^{-3})^4$
- h $(5 \times 10^{-3})^3$
- i $(6 \times 10^{-1}) \times (4 \times 10^3) \times (5 \times 10^{-4})$
- j $(6 \times 10^{-3})^2 \times (8 \times 10^{11})$
- k $(4 \times 10^3)^{-1}$
- l $(5 \times 10^{-4})^{-2}$

- 6 Simplify the following, writing your answers in scientific notation:

a $\frac{8 \times 10^6}{4 \times 10^3}$

b $\frac{9 \times 10^{-3}}{3 \times 10^{-1}}$

c $\frac{4 \times 10^6}{2 \times 10^{-2}}$

d $\frac{2.5 \times 10^{-4}}{(5 \times 10^7)^2}$

e $\frac{(8 \times 10^{-2})^2}{2 \times 10^{-6}}$

f $\frac{(5 \times 10^{-3})^{-2}}{(2 \times 10^4)^{-1}}$

- 7 a How many times larger is 3×10^{11} than 3×10^8 ?
- b i Which is smaller, 5×10^{-16} or 5×10^{-21} ?
- ii By how many times is it smaller than the other number?
- c How many times larger is 4×10^6 than 8×10^{-5} ?

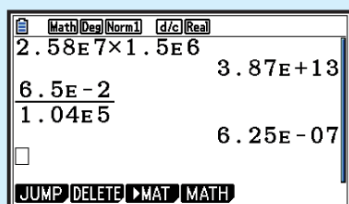
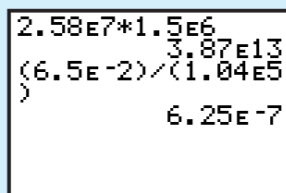
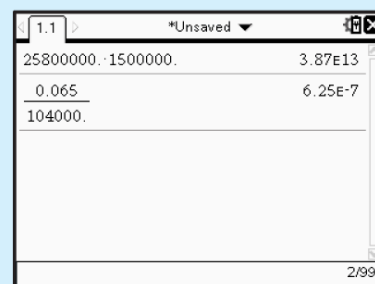
Example 9

Use your calculator to find:

a $(2.58 \times 10^7) \times (1.5 \times 10^6)$

b $\frac{6.5 \times 10^{-2}}{1.04 \times 10^5}$

Self Tutor

**GRAPHICS
CALCULATOR
INSTRUCTIONS**
Casio fx-CG20

TI-84 Plus

TI-nspire


a $(2.58 \times 10^7) \times (1.5 \times 10^6) = 3.87 \times 10^{13}$

b $\frac{6.5 \times 10^{-2}}{1.04 \times 10^5} = 6.25 \times 10^{-7}$

- 8** Calculate the following, giving each answer in scientific notation. The decimal part should be rounded to 3 significant figures.

a $(4.7 \times 10^5) \times (8.53 \times 10^7)$

b $(2.7 \times 10^{-3}) \times (9.6 \times 10^{14})$

c $\frac{3.4 \times 10^7}{4.8 \times 10^{15}}$

d $\frac{7.3 \times 10^{-7}}{1.5 \times 10^4}$

e $(2.83 \times 10^3)^2$

f $(5.96 \times 10^{-5})^2$

g $\frac{(3.56 \times 10^4)^2}{8.05 \times 10^{-5}}$

h $\frac{2.9 \times 10^2}{(7.62 \times 10^7)^3}$

- 9** Answer the **Opening Problem** on page 14.

- 10** Use your calculator to answer the following:

- a** A rocket travels in space at $4 \times 10^5 \text{ km h}^{-1}$. Assuming 1 year ≈ 365.25 days, how far will it travel in:

i 30 days

ii 20 years?

- b** A bullet travelling at an average speed of $2 \times 10^3 \text{ km h}^{-1}$ hits a target 500 m away. Find the time of the bullet's flight, in seconds.

- c** Mars has volume $1.31 \times 10^{21} \text{ m}^3$ whereas Pluto has volume $4.93 \times 10^{19} \text{ m}^3$.

How many times bigger is Mars than Pluto?

- d** Microbe C has mass 2.63×10^{-5} grams whereas microbe D has mass 8×10^{-7} grams.

i Which microbe is heavier?

ii How many times heavier is it, than the other microbe?


- 11** The table alongside shows the land areas of the Canadian provinces (shaded green) and territories (shaded purple).

- a** Find the total land area of Canada.
- b** Place the *provinces* in order, from largest to smallest.
- c** How many times larger is:
 - i** Quebec than Manitoba
 - ii** Nunavut than Prince Edward Island?
- d** What percentage of the land area of Canada, is included in Nova Scotia?

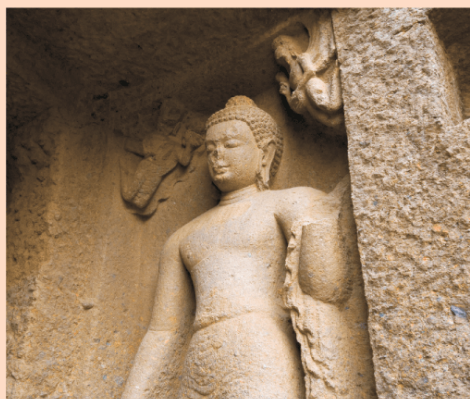
	Land area (km ²)
Ontario	9.2×10^5
Quebec	1.4×10^6
Nova Scotia	5.3×10^4
New Brunswick	7.1×10^4
Manitoba	5.5×10^5
British Columbia	9.3×10^5
Prince Edward Island	5.7×10^3
Saskatchewan	5.9×10^5
Alberta	6.4×10^5
Newfoundland and Labrador	3.7×10^5
Northwest Territories	1.2×10^6
Yukon	4.7×10^5
Nunavut	1.9×10^6

HISTORICAL NOTE

The ancient Indians explored the concept of expressing very large and very small numbers. In the *Lalitavistara Sutra*, a Sanskrit text dating from around the 4th century, it is written that the Buddha gave a description of the size of an atom.

In terms of the length of a finger bone, the Buddha stated that:

“.... each was the length of **seven grains of barley**, each of which was the length of **seven mustard seeds**, each of which was the length of **seven poppy seeds**, each of which was the length of **seven particles of dust stirred up by a cow**, each of which was the length of **seven specks of dust disturbed by a ram**, each of which was the length of **seven specks of dust stirred up by a hare**, each of which was the length of **seven specks of dust carried away by the wind**, each of which was the length of **seven tiny specks of dust**, each of which was the length of **seven minute specks of dust**, each of which was the length of **seven particles of the first atoms**.”



What to do:

- 1** Assuming a finger bone is 4 cm long, use the Buddha's description to estimate the length of an atom, in metres. Write your answer in scientific notation.
- 2** Research the size of a carbon atom. How accurate is the estimate in **1**?

REVIEW SET 1A

1 Simplify using the index laws:

a $k^5 \times k^3$

b $\frac{p^6}{p}$

c $(m^6)^8$

2 Remove the brackets of:

a $(3w)^2$

b $(2x^2y)^3$

c $\left(\frac{a}{b}\right)^6$

d $\left(\frac{1}{5n}\right)^3$

3 Simplify, giving answers in simplest rational form:

a 7^{-1}

b $\left(\frac{4}{3}\right)^{-1}$

c $11^0 - 11^{-1}$

d $\left(1\frac{3}{4}\right)^{-2}$

4 Write using scientific notation:

a 59 000

b 0.009

c 6 085 000

d 0.000 007 71

5 Evaluate without using a calculator:

a $49^{\frac{1}{2}}$

b $64^{-\frac{1}{3}}$

c $125^{\frac{4}{3}}$

d $27^{-\frac{2}{3}}$

6 Write as an ordinary decimal number:

a 6.23×10^5

b 3.008×10^{-4}

c 4.597×10^0

7 Write with a prime number base:

a $\sqrt[5]{16}$

b $\frac{1}{\sqrt[3]{9}}$

c $\frac{625}{\sqrt{5}}$

d $8\sqrt{32}$

8 Use your calculator to evaluate the following correct to 3 significant figures:

a $\sqrt[3]{20}$

b $\frac{1}{\sqrt[6]{100}}$

c $10^{\frac{5}{4}}$

d $15^{-\frac{3}{7}}$

9 Write without brackets or negative indices:

a $(mn)^{-2}$

b mn^{-2}

c $\left(\frac{x}{5y^2}\right)^{-3}$

10 Simplify the following, writing your answers in scientific notation:

a $(6 \times 10^5) \times (3 \times 10^6)$

b $(8 \times 10^9) \times (5 \times 10^{-4})$

c $\frac{9 \times 10^{-5}}{6 \times 10^3}$

d $(8 \times 10^7)^{-1}$

11 Write the answers to the following in scientific notation:

a The speed of light in a vacuum is about $2.998 \times 10^8 \text{ m s}^{-1}$.

Assuming 1 year ≈ 365.25 days, determine how far light travels in:

i 1 hour

ii 1 day

iii 1 year.

b How long does it take for light to travel:

i 1 m

ii 1 cm

iii 1 mm?

c In air, light travels at $2.989 \times 10^8 \text{ m s}^{-1}$ and sound travels at 343.2 m s^{-1} .
How many times faster is light than sound?

REVIEW SET 1B

1 Express in simplest form with a prime number base:

a 8^2

b $\frac{25^x}{125}$

c $\frac{49^{k+3}}{7^{k-1}}$

2 Simplify using one or more of the index laws:

a $5c^3 \times 3c^4$

b $\frac{14x^5y^2}{2x^2y}$

c $\left(\frac{3p}{q^{-3}}\right)^2$

3 Evaluate $27^{\frac{4}{3}}$ without using a calculator.

4 Write without brackets or negative indices: **a** $\left(\frac{a}{b}\right)^{-2}$ **b** $\left(\frac{3a^{-1}}{2b^2}\right)^{-3}$

5 **a** Write using index notation: **i** $\sqrt[3]{13}$ **ii** $\frac{1}{\sqrt[5]{40}}$

b Use your calculator to evaluate $\frac{1}{\sqrt[5]{40}}$ correct to 3 significant figures.

6 Simplify the following, writing your answers in scientific notation:

a $(7 \times 10^5) \times (3 \times 10^9)$

b $\frac{2.7 \times 10^{-4}}{4.5 \times 10^7}$

c $\frac{8 \times 10^7}{2 \times 10^{-3}}$

7 Write the following as powers of 2:

a $\sqrt[4]{8}$

b $\frac{1}{\sqrt[5]{16}}$

c $\frac{4}{\sqrt{32}}$

8 How many times larger is 3.5×10^{11} than 5×10^9 ?

9 Find the smaller of 2^{60} and 7^{20} without using a calculator.

10 The table alongside shows the diameters of the planets in the solar system.

a Find the diameter of Saturn in:

i kilometres **ii** centimetres.

b Find the radius of Venus.

c Write the planets in order of size, from smallest to largest.

d How many times greater is the diameter of:

i Uranus than Mercury

ii Jupiter than Mars?

Planet	Diameter
Mercury	4.88×10^6 m
Venus	1.21×10^7 m
Earth	1.27×10^7 m
Mars	6.79×10^6 m
Jupiter	1.40×10^8 m
Saturn	1.21×10^8 m
Uranus	5.11×10^7 m
Neptune	4.95×10^7 m

11 **a** Write $(\sqrt[5]{7} \times \sqrt[4]{7})^{20}$ as a power of 7.

b Hence, show that $\sqrt[5]{7} \times \sqrt[4]{7} = 7^{\frac{9}{20}}$.

c Use the index laws to show that $\sqrt[m]{a} \times \sqrt[n]{a} = a^{\frac{m+n}{mn}}$.

d Hence, write $\sqrt[3]{11} \times \sqrt[5]{11}$ as a power of 11.

ANSWERS

EXERCISE 1A

- 1 a 3^7 b x^9 c x^{5+n} d t^{12}
 e 7^4 f x^4 g t^{6-x} h t^{3m-1}
 i 5^6 j t^{12} k y^{3m} l a^{12m}
- 2 a 11^2 b 2^5 c 3^4 d 2^4
 e 5^4 f 7^{t+2} g 3^{a-2} h 2^{3p-2}
 i 7^2 j 3^{2-x} k 5^{4t} l 2^{3k-12}
 m 2^{2a-b} n 2^{3x-4y} o 5^{2x+4} p 3^6
- 3 a x^2y^2 b a^3b^3 c $x^2y^2z^2$ d $27b^3$ e $625a^4$
 f $100\,000x^5y^5$ g $\frac{p^2}{q^2}$ h $\frac{m^3}{n^3}$ i $\frac{x^4}{81}$
 j $\frac{125}{z^3}$ k $\frac{16a^4}{b^4}$ l $\frac{27x^3}{64y^3}$
- 4 a $8b^5$ b a^2b^2 c $6a^4b^2$ d $\frac{x^2y}{3}$
 e $\frac{a^6}{125b^3}$ f $\frac{8r^2}{5}$ g $16c^4d^3$ h $\frac{5k^2}{16}$
- 5 a 1 b $\frac{1}{6}$ c $\frac{1}{4}$ d 1 e 16 f $\frac{1}{16}$
 g 125 h $\frac{1}{125}$ i 49 j $\frac{1}{49}$ k 1000 l $\frac{1}{1000}$
- 6 a 1 b 1 c $2, t \neq 0$ d $1, t \neq 0$
 e 1 f 3 g $\frac{1}{25}$ h $\frac{1}{16}$ i $\frac{1}{x^5}$ j $\frac{8}{3}$
 k $\frac{3}{2}$ l 5 m 3 n $\frac{4}{5}$ o $\frac{11}{3}$ p 9
 q $\frac{27}{8}$ r $\frac{8}{27}$ s $\frac{25}{16}$ t $\frac{4}{25}$
- 7 a $\frac{1}{3b}$ b $\frac{3}{b}$ c $\frac{7}{a}$ d $\frac{1}{7a}$ e t^2
 f $\frac{y}{3x}$ g $\frac{1}{25t^2}$ h $\frac{t^2}{5}$ i $\frac{x}{y}$ j $\frac{1}{xy}$
 k $\frac{x}{y^3}$ l $\frac{1}{x^3y^3}$ m $\frac{1}{3pq}$ n $\frac{3}{pq}$ o $\frac{3p}{q}$
 p x^3y^5 q $\frac{125y^9}{x^6}$ r $\frac{4d^6}{c^2}$ s $\frac{r^6t^2}{9}$ t $\frac{125}{8p^3q^6}$
- 8 a $\frac{1}{a^{-n}} = \frac{a^0}{a^{-n}} = a^{0-(-n)} = a^{0+n} = a^n$ b $\left(\frac{a}{b}\right)^{-n} = \left[\left(\frac{a}{b}\right)^{-1}\right]^n = \left(\frac{b}{a}\right)^n = \frac{b^n}{a^n}$
- 9 a ms^{-1} b m^3h^{-1} c cm^2s^{-1} d $\text{cm}^3\text{min}^{-1}$
 e gs^{-1} f kg ms^{-1} g ms^{-2}
- 10 $2^{125} = (2^5)^{25}$ and $3^{75} = (3^3)^{25}$
 $= 32^{25}$ $= 27^{25}$
 As $27^{25} < 32^{25}$, 3^{75} is smaller.

- 11 The indices have a common factor of 6.

$$2^{90} = (2^{15})^6 = 32\,768^6$$

$$3^{60} = (3^{10})^6 = 59\,049^6$$

$$5^{36} = (5^6)^6 = 15\,625^6$$

$$10^{24} = (10^4)^6 = 10\,000^6$$

$$\text{Order: } 10^{24}, 5^{36}, 2^{90}, 3^{60}$$

EXERCISE 1B

- 1 a 4 b $\frac{1}{4}$ c 5 d $\frac{1}{5}$ e 2 f $\frac{1}{2}$
 g -2 h $-\frac{1}{2}$ i 3 j $\frac{1}{3}$ k 2 l $\frac{1}{2}$
- 2 a not possible b -1 c $-\frac{1}{3}$ d not possible
- 3 a $10^{\frac{1}{2}}$ b $10^{-\frac{1}{2}}$ c $15^{\frac{1}{3}}$ d $15^{-\frac{1}{3}}$
 e $19^{\frac{1}{4}}$ f $19^{-\frac{1}{4}}$ g $13^{\frac{1}{5}}$ h $13^{-\frac{1}{5}}$
- 4 a 4 b 8 c 32 d 32 e 8 f 27
 g $\frac{1}{27}$ h $\frac{1}{2}$ i 2 j 4 k 8 l $\frac{1}{8}$
 m $\frac{1}{4}$ n $\frac{1}{81}$ o $\frac{1}{125}$
- 5 a $2^{\frac{3}{2}}$ b $2^{\frac{5}{3}}$ c $2^{\frac{7}{2}}$ d $2^{\frac{4}{3}}$ e $2^{-\frac{3}{4}}$ f $2^{-\frac{4}{3}}$
 g $2^{-\frac{3}{7}}$ h $2^{-\frac{6}{5}}$ i $2^{\frac{7}{2}}$ j $2^{\frac{9}{2}}$ k $2^{\frac{1}{3}}$ l $2^{\frac{2}{3}}$
- 6 a $3^{\frac{2}{3}}$ b $3^{\frac{3}{2}}$ c $3^{-\frac{3}{4}}$ d $3^{-\frac{4}{5}}$ e $3^{\frac{5}{2}}$ f $3^{\frac{5}{2}}$
 g $3^{\frac{9}{5}}$ h $3^{-\frac{2}{3}}$
- 7 a $5^{\frac{2}{3}}$ b $2^{\frac{5}{4}}$ c $5^{\frac{3}{5}}$ d $11^{\frac{2}{7}}$ e $7^{-\frac{2}{3}}$ f $2^{\frac{6}{5}}$
 g $5^{-\frac{4}{7}}$ h $3^{-\frac{5}{6}}$ i $2^{\frac{11}{2}}$ j $5^{\frac{7}{2}}$ k $13^{\frac{1}{3}}$ l $3^{\frac{5}{2}}$
- 8 a 125 b 9 c 128 d ≈ 2.41
 e ≈ 2.68 f ≈ 91.2 g ≈ 1.93 h ≈ 849
 i ≈ 2.13 j ≈ 9.16 k 0.0313 l $\approx 0.004\,12$
 m ≈ 0.339 n ≈ 0.182 o ≈ 0.215
- 9 3

EXERCISE 1C

- 1 a 2.3×10^2 b 5.39×10^4 c 3.61×10^{-2}
 d 6.8×10^{-3} e 3.26×10^0 f 5.821×10^{-1}
 g 3.61×10^8 h 1.674×10^{-6}
- 2 a 2300 b 0.023 c 564 000
 d 0.000 793 1 e 9.97 f 60 400 000
 g 0.4215 h 0.000 000 036 21
- 3 a $\approx 4 \times 10^6$ red blood cells b 8×10^{-4} m
 c 6.38×10^6 m d 4.3252×10^{19} arrangements
- 4 a 6 990 000 m b 0.018 cm
 c 32 000 000 bacteria d 0.000 008 2 t
- 5 a 6×10^{10} b 2.8×10^9 c 5.6×10^{-8}
 d 5.4×10^{-6} e 9×10^{10} f 1.6×10^{15}
 g 1.6×10^{-11} h 1.25×10^{-7} i 1.2×10^0
 j 2.88×10^7 k 2.5×10^{-4} l 4×10^6
- 6 a 2×10^3 b 3×10^{-2} c 2×10^8
 d 1×10^{-19} e 3.2×10^3 f 8×10^8
- 7 a 1000 times b i 5×10^{-21} ii 100 000 times
 c 5×10^{10} times
- 8 a $\approx 4.01 \times 10^{13}$ b $\approx 2.59 \times 10^{12}$ c $\approx 7.08 \times 10^{-9}$
 d $\approx 4.87 \times 10^{-11}$ e $\approx 8.01 \times 10^6$ f $\approx 3.55 \times 10^{-9}$
 g $\approx 1.57 \times 10^{13}$ h $\approx 6.55 \times 10^{-22}$
- 9 a 5.84×10^7 b i 1.60×10^5 ii 5.84×10^8

- 10 a i 2.88×10^8 km ii $\approx 7.01 \times 10^{10}$ km b 0.9 s
c ≈ 26.6 times d i Microbe C ii ≈ 32.9 times
- 11 As land areas are rounded to 2 significant figures, the answers should be rounded to 2 significant figures.
a 9.1×10^6 km²
b Quebec > British Columbia > Ontario > Alberta > Saskatchewan > Manitoba > Newfoundland and Labrador > New Brunswick > Nova Scotia > Prince Edward Island
c i ≈ 2.5 times ii ≈ 330 times d $\approx 0.58\%$

REVIEW SET 1A

- 1 a k^8 b p^5 c m^{48}
- 2 a $9w^2$ b $8x^6y^3$ c $\frac{a^6}{b^6}$ d $\frac{1}{125n^3}$
- 3 a $\frac{1}{7}$ b $\frac{3}{4}$ c $\frac{10}{11}$ d $\frac{16}{49}$
- 4 a 5.9×10^4 b 9×10^{-3} c 6.085×10^6 d 7.71×10^{-6}
- 5 a 7 b $\frac{1}{4}$ c 625 d $\frac{1}{9}$
- 6 a 623 000 b 0.000 300 8 c 4.597
- 7 a $2^{\frac{4}{5}}$ b $3^{-\frac{2}{3}}$ c $5^{\frac{7}{2}}$ d $2^{\frac{11}{2}}$
- 8 a ≈ 2.71 b ≈ 0.464 c ≈ 17.8 d ≈ 0.313
- 9 a $\frac{1}{m^2n^2}$ b $\frac{m}{n^2}$ c $\frac{125y^6}{x^3}$
- 10 a 1.8×10^{12} b 4×10^6 c 1.5×10^{-8} d 1.25×10^{-8}
- 11 a i $\approx 1.079 \times 10^{12}$ m ii $\approx 2.590 \times 10^{13}$ m
iii $\approx 9.461 \times 10^{15}$ m
b i $\approx 3.336 \times 10^{-9}$ s ii $\approx 3.336 \times 10^{-11}$ s
iii $\approx 3.336 \times 10^{-12}$ s
c $\approx 8.71 \times 10^5$ times faster

REVIEW SET 1B

- 1 a 2^6 b 5^{2x-3} c 7^{k+7}
- 2 a $15c^7$ b $7x^3y$ c $9p^2q^6$ d 81
- 4 a $\frac{b^2}{a^2}$ b $\frac{8a^3b^6}{27}$
- 5 a i $13^{\frac{1}{3}}$ ii $40^{-\frac{1}{5}}$ b ≈ 0.478
- 6 a 2.1×10^{15} b 6×10^{-12} c 4×10^{10}
- 7 a $2^{\frac{3}{4}}$ b $2^{-\frac{4}{5}}$ c $2^{-\frac{1}{2}}$ d 70 times
- 9 7^{20} as $2^{60} = (2^3)^{20} = 8^{20}$ and $8^{20} > 7^{20}$
- 10 a i 1.21×10^5 km ii 1.21×10^{10} cm
b 6.05×10^6 m
c Mercury < Mars < Venus < Earth < Neptune < Uranus < Saturn < Jupiter
d i ≈ 10.5 times larger ii 20.6 times larger
- 11 a 7^9 b As $(\sqrt[5]{7} \times \sqrt[4]{7})^{20} = 7^9$ d $\sqrt[3]{11} \times \sqrt[5]{11}$
then $\sqrt[5]{7} \times \sqrt[4]{7} = 7^{\frac{9}{20}} = 11^{\frac{3+5}{3 \times 5}} = 11^{\frac{8}{15}}$

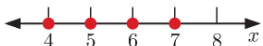
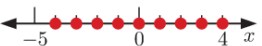
EXERCISE 2A

- 1 a $8 \in P$ b $k \notin S$ c $14 \notin \{\text{odd numbers}\}$
d $n(Y) = 9$
- 2 a true b true c true d true
e false f false g true h true
- 3 a rational b rational c rational d rational
e irrational f neither g rational h rational
i neither j rational

- 4 a i $A = \{1, 2, 3, 6\}$ ii finite iii $n(A) = 4$
b i $B = \{6, 12, 18, 24, \dots\}$ ii infinite
c i $C = \{1, 17\}$ ii finite iii $n(C) = 2$
d i $D = \{17, 34, 51, 68, \dots\}$ ii infinite
e i $E = \{2, 3, 5, 7, 11, 13, 17, 19\}$ ii finite
iii $n(E) = 8$
f i $F = \{12, 14, 15, 16, 18, 20, 21, 22, 24, 25, 26, 27, 28\}$
ii finite iii $n(F) = 13$

- 5 a $0.\overline{7} = \frac{7}{9}$ b $0.\overline{41} = \frac{41}{99}$ c $0.\overline{324} = \frac{12}{37}$
- 6 0.527 can be written as $\frac{527}{1000}$, where 527 and 1000 are integers
- 7 $0.\overline{9} = 1$ and $1 \in \mathbb{Z}$
- 8 Note: There may be other answers.
a $\sqrt{2}$, $-\sqrt{2}$ are irrational, but $\sqrt{2} + (-\sqrt{2}) = 0$ which is rational.
b $\sqrt{2}$, $\sqrt{50}$ are irrational, but $\sqrt{2} \times \sqrt{50} = \sqrt{100} = 10$ which is rational.

EXERCISE 2B

- 1 a The set of all real x such that x is greater than 4.
b The set of all integers x such that x is less than or equal to 5.
c The set of all real y such that y lies between 0 and 8.
d The set of all integers x such that x lies between 1 and 4 including 1 and 4.
e The set of all real t such that t lies between 2 and 7.
f The set of all real n such that n is less than or equal to 3 or n is greater than 6.
- 2 a $\{x \mid x > 3\}$ b $\{x \mid 2 < x \leq 5\}$
c $\{x \mid x \leq -1 \text{ or } x \geq 2\}$ d $\{x \mid -1 \leq x < 5, x \in \mathbb{Z}\}$
e $\{x \mid 0 \leq x \leq 6, x \in \mathbb{N}\}$ f $\{x \mid x < 0\}$
- 3 a  b 
c  d 
e  f 
- 4 a $\{x \mid x > 7\}$ b $\{x \mid -8 < x < 15, x \in \mathbb{Z}\}$
c $\{x \mid 4 \leq x < 6, x \in \mathbb{Q}\}$

EXERCISE 2C

- 1 a $A \subseteq B$ b $A \not\subseteq B$ c $A \subseteq B$ d $A \not\subseteq B$
e $A \subseteq B$
- 2 a true b true c false d true e false f true
- 3 $B \not\subseteq A$, $C \subseteq A$
- 4 a false b true c false d true
- 5 a $A' = \{1, 3, 4, 7, 8, 9\}$ b $B' = \{1, 4, 6, 8, 9\}$
c $C' = \{2, 4, 6, 8\}$ d $D' = \{1, 2, 3, 5, 6, 7, 9\}$
e $E' = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ (or $E' = U$)
f $F' = \{3, 4, 5, 6, 7, 8, 9\}$
- 6 a $P' = \{A, B, D, E, G, H, I, K, L, N, O, Q, R, S, T, V, W, X\}$
b $Q' = \{\text{vowels}\}$
c $R' = \{B, C, D, E, F, G, J, K, M, N, Q, R, U, V, W, X, Y, Z\}$
d $S' = \{A, B, C, D, E, F, G, H, I, J\}$